

# Statistics

## Parameter Estimation, Hypothesis Testing, Regression

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- Thus statistics is about making inferences from the sample to the population.

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  - To summarise: Statistics are what the investigator **knows** from the samples. Parameters are what they **want** to know.

# Example application: Periodic Labour Force Survey

Wednesday, March 25, 2026 | 05:42 PM IST



## Business Standard

Over two-thirds of unemployed Indians aged 20–29 were graduates in 2023, with the share of degree holders among the unemployed rising sharply from 46 per cent in 2017 to 67 per cent in 2023, according to the State of Working India 2026 report released by Azim Premji University on Tuesday.

The report also found that less than half of all graduates were employed, with 48.8 per cent reporting some form of work. Among graduates, only 6.7 per cent had permanent salaried jobs, while 3.7 per cent were in white-collar occupations.

“Between 2004–05 and 2023, while roughly 5 million graduates were added each year, only around 2.8 million found employment, and an even smaller share entered salaried jobs, contributing to rising graduate unemployment and slowing earnings growth,” the report observed.

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  - Twelve households are selected in each FSU. 272,304 households are queried for data.
  - Convert survey into numbers for India.

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- Television rating point (TRP) is used in determining the reach and demographic.

## Hindi TV Serials TRP & FMN Rating

### Top 5 TV Serials Of The Week

Week 10, 2026

| Rank | Name                             | TRP | Channel   |
|------|----------------------------------|-----|-----------|
| 1    | Kyunki Saas Bhi Kabhi Bahu Thi 2 | 2.0 | Star Plus |
| 2    | Anupamaa                         | 1.9 | Star Plus |
| 3    | Tum Se Tum Tak                   | 1.8 | Zee TV    |
| 4    | Taarak Mehta Ka Ooltah Chasmah   | 1.8 | Sab TV    |
| 5    | Vasudha                          | 1.7 | Zee TV    |

**TRP में फर्जीवाड़ा:** रिपब्लिक विवाद के बाद ब्रॉडकास्ट काउंसिल BARC ने न्यूज चैनलों की वीकली TRP लिस्ट पर अस्थायी रोक लगाई

मुंबई 6 दिन पहले

## CBI Moves To Investigate Fake Ratings Case As Complaint Is Filed In UP

Republic TV has been demanding a CBI probe, accusing the Mumbai police of going after their recent clash over the Sushant Singh Rajput case.

— reported by Arvind Gunasekar, Sreenivasan Jain, Edited by Deepshikha Ghosh | Updated: October 21, 2020 7:07 am IST

## The Manipulation of Television Rating Points: How TRPs work, the scam

Mumbai police are looking at alleged manipulation of Television Rating Points. How do channels score TRPs? What are the ways in which manipulation is possible, and how often has it been alleged in the past?

**TRP scam case: BARC says 'disappointed with Republic TV for disclosing, misrepresenting' confidential communication**

The statement comes after the Republic Media Network earlier in the day claimed that BARC, in an email said it had found no complaint or evidence of malpractice against Republic TV, RepublicBharat or any of its affiliates

TRP scam | October 18, 2020 21:28:33 IST

## TRP Ratings Scam: CBI Registers Case to Probe Alleged Fraud Based on Complaint Filed in UP

The CBI probe into the case comes amid an ongoing investigation by the Mumbai Police. Several television news channels are under the latter's radar, including the Republic TV headed by Arnab Goswami.



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- *Reach* and *Target audience*—*Gross Rating Point* and *Target Rating Point* are used by advertisers to buy ad time.

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- In India, TRP measurement is conducted by several organisations—Joint Industry Board Television Audience Measurement Research (JIB-TAM), ORG-MARG’s Indian Television Audience Measurement (INTAM), (these two have now merged), Audience Measurement and Analytics (aMap), Doordarshan Audience Ratings team (DART);

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- Broadcast Audience Research Council (BARC) is a cross industry body that oversees the TV audience measurement. Telecom Regulatory Authority of India (TRAI) regulates the process and provides guidelines for operation.

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  - The rural population is not covered at all! Rural viewing profile is very different from urban.
  - DTH is not covered.
- Data is collected overnight and the rating published on Friday.

# TRP ratings: some methodology

- TAM is conducted in about 60,000 homes that have cable TV service by placing “people meters” in these homes.
- Channels that authorise BARC to measure their TRP rating, will include an audio watermark. People meter uses this to record the details of viewing by the family.
- Selecting the households to meter is a complex process: rural vs urban (megacities and Tier II), HSM vs South, public vs private, etc are the various criteria that will be used to select the household.
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- Some obvious problems:
  - The rural population is not covered at all! Rural viewing profile is very different from urban.
  - DTH is not covered.
- Data is collected overnight and the rating published on Friday. A lot of cleaning is required.
- TRP rating determines the price of advertising in the slot.

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- Secrecy: The broadcasters and the cable TV suppliers should not know the families that are part of the survey.
- Effect of chasing TRP rating: Even the definition of news changes—entertainment and crime is more than 30% of the news!

# TRP ratings: some more economics

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<https://www.barcindia.co.in/measurement/television-au>

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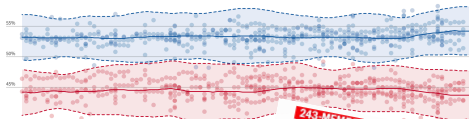
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- And how bad: In the 1948 US Presidential election (Harry Truman vs Thomas Dewey) three separate polls said Dewey would get 50, 50, and 53 % of the vote and Truman would get 45, 44, and 38% of the vote. Actual result: 50% for Truman and 45% for Dewey! And here is the famous picture.

## US presidential election 2020: is Donald Trump or Joe Biden leading in our poll tracker?

Last updated: October 21

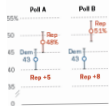
See how the opinion polls have shifted over the course of the election campaign. Each dot represents the result of a published voting intention survey while the solid lines represent our weighted polling average and the dotted lines show margin of error.



For election polls, different measures of the race have different margins of error

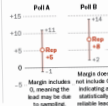
The margin of error reported for most polls applies to support for individual candidates ...

Margin of error for single candidate support (MOE +/- 3 pct. points)

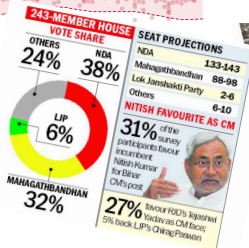


... while the margin of error for a candidate's lead is nearly twice as large.

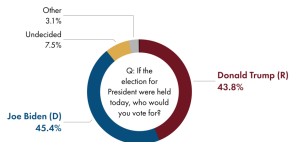
Margin of error for difference between two candidates' level of support (SDep +/- 6 pct. points)



Source: Hypothetical polling results from a two-party election.  
PEW RESEARCH CENTER



## Trump vs. Biden spread within the margin of error in Arizona



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- Regression: propose a relation between a set of explanatory variables and ‘response variable’ determine parameters of the model and/or test hypotheses about the relation.
- Key concepts to be used: pdf and pmf, Independent random variables, moments, CLT and LLN to provide some intuition.